pivot

arr = [8, 1, 3, $\boxed{4}$, 20, 50, 30]

Partition logic ⇒ (A) pivot choose ⇒ pivot index = Start or end or any ele.

(B) Place the pivot to it's correct position

↳ count no < pivot Element
{ swap (arr[pivotIndex], arr[s + count])
}

(C) left mai sare chote ele pivot se
Right mai sare bade ele pivot se

[4, 1, 3, $\boxed{8}$, 20, 50, 30]
left right

(4) Recursive call

left → | 4 | 1 | 3 | (0 1 2)

Right → | 20 | 50 | 30 | (4 5 6) ← e, s

| 4 | 1 | 3 | 8 | 20 | 50 | 30 | (i, j)

```
while ( i < pivotIndex && j < pivotIndex)
{       while (arr[i] ≤ pivotElement)
        {       i++;
        }
        while ( arr[j] > pivotElement)
        {       j--;
        }
```

if any case where we found element with wrong position

eg → [4, 3, 40, $\boxed{10}$, 7, 30, 50]

→ check if it is in range

```
if ( i < pivotIndex && j > pivotIndex)
{       swap( arr[i] , arr[j] )
}
```